

# UC300 HS24 Cat.5e LSHF-FR D<sub>ca</sub>

## SF/UTP Installation Cable



## Application

Primary (Campus), Secondary (Riser), Tertiary (Horizontal)  
IEEE 802.3: 10Base-T; 100Base-T; 1000Base-T;  
IEEE 802.5 16 MB; ISDN; TPDDI; ATM  
Power over Ethernet (PoE) / Type 1-4

## Standards

EN 50173-1; EN 50288-2-1  
ISO/IEC 11801; IEC 61156-5  
TIA/EIA-568-C.2  
IEEE 802.3af / at / bt

## Flame resistance

EN 50399: Class D<sub>cas</sub>2d2a1;  
LSHF (LSOH): IEC 60332-3-24; IEC 60332-1; IEC 60754-2; IEC 61034

## Construction

|              |                                                                             |
|--------------|-----------------------------------------------------------------------------|
| Conductor    | bare copper wire, Ø 0.51 mm (AWG24)                                         |
| Insulation   | PE, Ø 1.1 mm                                                                |
| Twisting     | 2 cores to the pair                                                         |
| Cable lay up | 4 pairs to the core                                                         |
| Screen       | Al-laminated plastic foil and Copper braid, tinned                          |
| Sheath       | LSHF-FR Dca, grey RAL 7035<br>Duplex sheath: two cables parallel, separable |

## Mechanical properties

|                   |                     |                 |
|-------------------|---------------------|-----------------|
| Bending radius    | without load        | 4 x D           |
|                   | with load           | 8 x D           |
| Temperature range | during operation    | -20°C to + 60°C |
|                   | during installation | 0°C to + 50°C   |

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### Electrical properties

at 20°C ± 5°C

|                                 |                                       |                |
|---------------------------------|---------------------------------------|----------------|
| Loop resistance                 |                                       | ≤ 190 Ω/km     |
| Resistance unbalance            |                                       | ≤ 2%           |
| Insulation resistance           | (500 V)                               | ≥ 2000 MΩxkm   |
| Mutual capacitance              | at 800 Hz                             | Nom. 48 nF/km  |
| Capacitance unbalance           | (pair/ground)                         | ≤ 1500 pF/km   |
| Mean characteristic impedance   | 100 MHz                               | 100 ± 5 Ω      |
| Nominal velocity of propagation |                                       | ca. 67%        |
| Propagation delay               |                                       | ≤ 535 ns/100 m |
| Delay skew                      |                                       | ≤ 20 ns/100 m  |
| Test voltage                    | (DC, 1 min) core/core and core/screen | 1000 V         |
| Transfer impedance              | at 1 MHz                              | ≤ 20 mΩ /m     |
|                                 | at 10 MHz                             | ≤ 20 mΩ /m     |
|                                 | at 30 MHz                             | ≤ 30 mΩ /m     |
|                                 | at 100MHz                             | ≤ 60 mΩ /m     |
| Coupling attenuation            |                                       | ≥ 75 dB        |
| Segregation class               |                                       | "c"            |

### Electrical data (nominal)

acc. to Cat.5e (at 20°C)

| F<br>(MHz) | Attenuation<br>(dB/100m) | NEXT<br>(dB) | PS-NEXT<br>(dB) | ACR<br>(dB/100m) | PS-ACR<br>(dB/100m) | ELFEXT<br>(dB/100m) | PS-ELFEXT<br>(dB/100m) | Return<br>loss<br>(dB) |
|------------|--------------------------|--------------|-----------------|------------------|---------------------|---------------------|------------------------|------------------------|
| 1.0        | 1.9                      | 71           | 68              | 69.1             | 66.1                | 68                  | 65                     | 20                     |
| 4.0        | 3.7                      | 62           | 59              | 58.3             | 55.3                | 56                  | 53                     | 23                     |
| 10.0       | 6.0                      | 56           | 53              | 50.0             | 47.0                | 48                  | 45                     | 25                     |
| 16.0       | 7.6                      | 53           | 50              | 45.4             | 42.4                | 44                  | 41                     | 25                     |
| 20.0       | 8.5                      | 51           | 48              | 42.5             | 39.5                | 42                  | 39                     | 25                     |
| 31.2       | 10.7                     | 49           | 46              | 38.3             | 35.3                | 38                  | 35                     | 24                     |
| 62.5       | 15.7                     | 44           | 41              | 28.3             | 25.3                | 32                  | 29                     | 22                     |
| 100.0      | 19.8                     | 41           | 38              | 21.2             | 18.2                | 28                  | 25                     | 20                     |
| 125.0      | 22.3                     | 40           | 37              | 17.7             | 14.7                | 26                  | 23                     | 19                     |
| 155.5      | 24.2                     | 38           | 35              | 13.8             | 10.8                | 24                  | 21                     |                        |
| 175.0      | 25.7                     | 37           | 34              | 11.3             | 8.3                 | 23                  | 20                     |                        |
| 200.0      | 27.5                     | 36           | 33              | 8.5              | 5.5                 | 22                  | 19                     |                        |
| 250.0      | 29.2                     | 35           | 32              | 5.8              | 2.8                 | 20                  | 17                     |                        |
| 300.0      | 32.0                     | 34           | 31              | 2.0              | -1.0                | 16                  | 13                     |                        |

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### Product order data

| Product reference code | Cable type                 | Brand name                                                        | Outer diameter | Euro Class             | Fire load |       | Weight | Cu Content | Tensile force |
|------------------------|----------------------------|-------------------------------------------------------------------|----------------|------------------------|-----------|-------|--------|------------|---------------|
|                        |                            |                                                                   | mm             |                        | MJ/km     | kWh/m |        |            | N             |
| 60061333               | J-2Y(St)CH<br>4x2x0.51     | UC300 HS24 Cat.5e<br>SF/UTP 4P LSHF-FR D <sub>ca</sub>            | 6.8            | D <sub>ca</sub> s2d2a1 | 433       | 0.120 | 47     | 27         | 120           |
| 60064235               |                            | UC300 HS24 Cat.5e<br>SF/UTP 4P LSHF-FR D <sub>ca</sub><br>500DP   |                |                        |           |       |        |            |               |
| 60061334               |                            | UC300 HS24 Cat.5e<br>SF/UTP 4P LSHF-FR D <sub>ca</sub><br>1000DW  |                |                        |           |       |        |            |               |
| 60061335               | J-2Y(St)CH<br>2x(4x2x0.51) | UC300 HS24 Cat.5e<br>SF/UTP 2x4P LSHF-FR D <sub>ca</sub>          | 6.8/14.8       | D <sub>ca</sub> s2d2a1 | 866       | 0.240 | 115    | 54         | 240           |
| 60061336               |                            | UC300 HS24 Cat.5e<br>SF/UTP 2x4P LSHF-FR D <sub>ca</sub><br>500DW |                |                        |           |       |        |            |               |

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